

ASCEND INSTITUTE, GANGTOK
CHAPTER-WISE TEST
CBSE CLASS 6 - MATHEMATICS

Chapter: Prime Numbers

Date: _____

Name: _____

Maximum Marks: 20

Time: 40 minutes

Instructions: Answer all questions. Section A contains 8 MCQs of 1 mark each. Section B contains 6 questions of 2 marks each. Show the necessary steps wherever required.

SECTION A - 8 MARKS

1. Which of the following is a prime number?

- | | |
|--------|--------|
| (i) 51 | (j) 57 |
| (k) 59 | (l) 69 |

Answer: _____

2. Which number is composite?

- | | |
|--------|--------|
| (i) 29 | (j) 31 |
| (k) 37 | (l) 39 |

Answer: _____

3. The prime factorisation of 84 is:

- | | |
|------------------------------------|----------------------------|
| (i) $2 \times 2 \times 3 \times 7$ | (j) $2 \times 3 \times 14$ |
| (k) 4×21 | (l) 2×42 |

Answer: _____

4. Which pair is co-prime?

- | | |
|---------------|---------------|
| (i) 8 and 15 | (j) 6 and 9 |
| (k) 10 and 15 | (l) 12 and 18 |

Answer: _____

5. Which is the smallest prime number?

- | | |
|-------|-------|
| (i) 0 | (j) 1 |
| (k) 2 | (l) 3 |

Answer: _____

6. How many prime numbers are there between 10 and 20?

- | | |
|-------|-------|
| (i) 2 | (j) 3 |
| (k) 4 | (l) 5 |

Answer: _____

7. Which number has exactly three prime factors when repeated factors are counted?

- | | |
|--------|--------|
| (i) 12 | (j) 13 |
| (k) 17 | (l) 19 |

Answer: _____

8. Which statement is true?

- | | |
|-------------------------------|-------------------------------------|
| (i) 1 is a prime number | (j) 2 is the only even prime number |
| (k) Every odd number is prime | (l) Every prime number is odd |

Answer: _____

SECTION B - 12 MARKS

Answer all questions. Continue on the back side if additional space is required.

9. Express 90 as a product of prime factors. [2]

10. Is 91 prime or composite? Write a factor pair to support your answer. [2]

11. List all prime numbers between 30 and 50. [2]

12. Express 180 as a product of prime factors. [2]

13. Write any two pairs of twin primes less than 30. [2]

14. Find the common prime factors of 60 and 84. [2]
